

FLAPPY BIRD GAME

SYNOPSIS REPORT OF MINI PROJECT

(Flappy bird game using python)

In partial fulfillment of requirements for the degree of

BACHELOR OF TECHNOLOGY

IN

**COMPUTER SCIENCE AND ENGINEERING
(INTERNET OF THINGS)**

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INTRODUCTION

“Flappy bird, a mobile gaming sensation introduced in 2013 by Vietnamese developer Dong Nguyen, is a deceptively simple et highly addictive game. In this side-scrolling 2D arcade game, players control a small, pixelated bird by tapping on the screen, causing the bird to flap its wings and ascend.

The objective is to navigate through a series of green pipes that resemble those from Nintendo’s Mario franchise. The challenge lies in the precise timing required to navigate the bird through openings in the pipes, as a single collision results in game over.

What sets Flappy Bird apart is its minimalist design – featuring basic graphics and a straightforward interface. Despite its apparent simplicity, the game proved to be infuriatingly difficult, captivating players worldwide with its challenging mechanics and encouraging them to compete for high scores. The game’s difficulty, combined with its addictive nature, led to a rapid rise in popularity.

Flappy Bird’s rise to fame was meteoric, propelled by its addictive nature and the burgeoning social media era. Players shared their high scores, vented their frustrations, and engaged in a collective experience that transcended cultural boundaries. The game’s minimalist design, featuring a single, looping background and simplistic obstacles, contributed to its accessibility and widespread appeal.

Flappy Bird’s success, however, was accompanied by controversy. The game’s creator, Dong Nguyen, faced immense pressure and criticism due to its addictive qualities and the impact on players. In a surprising move, Nguyen eventually decided to remove Flappy Bird from app stores in 2014, citing concerns about the game’s addictive nature and the overwhelming attention it brought.

Despite its brief existence, Flappy Bird left an incredible mark on the mobile gaming landscape, showcasing the unpredictable dynamics of the industry and the power of simplicity in captivating a global audience.

REQUIREMENT SPECIFICATION

MINIMUM REQUIREMENTS:

- **Operating system:** Android 4.1 or iOS 8.0 and above.
- **Processor:** 1 GHz or equivalent.
- **RAM:** 512 MB.
- **Storage:** 50 MB of available space.
- **Graphics:** Basic graphics support.

RECOMMENDED REQUIREMENTS:

- **Operating system:** Android 9.0 or iOS 14.0 and above.
- **Processor:** Dual-core 1.5 GHz or equivalent.
- **RAM:** 1 GB
- **Storage:** 100 MB of available space.
- **Graphics:** Accelerated graphics with OpenGL ES 2.0 support.
- **Display:** 720*1280 pixels resolution or higher.

NOTE: Flappy Bird is a lightweighted game, and the recommended requirements aim to enhance the overall gaming experience by ensuring smoother performance and improved on newer devices.

FUTURE SCOPE

The future scope of Flappy Bird envisions strategic enhancements and adaptations to meet evolving gaming trends:

1. **INTEGRATION OF AUGMENTED REALITY(AR):** Incorporating AR features could transform Flappy Bird into an interactive and immersive experience. Players might navigate the bird through real-world environments, adding a dynamic layer to the gameplay.
2. **MULTIPLAYER MODES AND SOCIAL INTEGRATION:** Introducing multiplayer modes and social gaming elements can create a sense of community. Players could compete in real-time, share achievements, and engage in collaborative challenges, fostering a more connected player base.
3. **CROSS-PLATFORM COMPATIBILITY:** Expanding the game to different platforms, including virtual reality (VR) headsets or gaming consoles, could cater to a broader audience. This would allow players to experience Flappy Bird in various ways, adapting to the capabilities of different devices.
4. **DYNAMIC DIFFICULTY ADJUSTMENT:** Implementing machine learning algorithms to adapt the game's difficulty based on individual player skills would personalize the gaming experience. This ensures that both novice and experienced players find the game challenging and enjoyable.
5. **THEMED VARIATIONS AND SEASONAL UPDATES:** Introducing themed variations of Flappy Bird tied to popular culture or seasonal events keeps the game fresh. Seasonal challenges, special events, and limited-time content contribute to sustained player engagement.
6. **COLLABORATIONS AND CUSTOMIZATION:** Exploring partnerships with other franchises or allowing players to customize their gaming experience with skins, backgrounds, or additional features adds a layer of personalization. This not only attracts new players but also retains the interest of existing ones.
7. **CONTINUOUS INNOVATION:** Regular updates with new obstacles, power-ups, or game modes demonstrate a commitment to innovation. Embracing player feedback and incorporating popular suggestions can keep Flappy Bird relevant in the competitive mobile gaming landscape.

By strategically adopting these elements, the future of Flappy Bird could involve a seamless blend of technology, social interaction, and continual innovation, ensuring its enduring popularity and relevance in the gaming industry.

INPUT & OUTPUT

Input:

1. **Tap or click Input:** The primary input is the player's tap or click on the screen, causing the bird to flap its wings and ascend.

Output:

1. **Bird Movement:** The bird ascends when the player taps or clicks and descends due to gravity when there's no input.
2. **Collision Detection:** If the bird collides with the pipes or the ground, the game ends, and a game-over screen is displayed.
3. **Score Display:** The score increments as the bird successfully passes through pipes, displayed on the screen during gameplay.
4. **Game Over Screen:** When the game ends, a screen shows the player's final score and an option to restart the game.